VR Lab Scenario

Analysis & Design Document

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Project: Virtual Reality Lab for Training & Research

Client: Gamyra

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a. Understanding of the Client's Needs

The client aims to establish an immersive VR Lab focused on both **training** and **research**. The solution should allow users—particularly those new to VR—to explore a **virtual lab environment**, interact with **virtual objects**, and be **guided by a smart assistant**. The assistant should offer onboarding and continuous support throughout the session.

Key functionalities:

- A **virtual lab space** that is intuitive and easy to navigate.
- **User movement and rotation** using the VR joystick to look around and explore the environment naturally.
- **Object interaction**, including grabbing, dragging, and placing virtual items into designated areas.
- A **smart voice assistant** that guides the user during onboarding and throughout the experience with contextual voice prompts.
- **User session saving/loading** to allow researchers to analyze progress, repeat scenarios, or resume sessions.
- A **progress tracking system** that logs user actions, completion status, and timestamps to support research analysis.

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b. Key Assumptions Made

1. **User Base:** Users are unfamiliar or semi-familiar with VR. The design must be intuitive.
2. **Hardware:** The lab will use VR headsets with controllers supporting joystick navigation and button interactions.
3. **Session Data:** Each session needs to be saved in a structured format (e.g., JSON/relational DB) for researchers to evaluate later.
4. **Smart Assistant:** Pre-recorded voice clips or synthesized voice instructions will guide users.
5. **Training Modules:** The lab experience is modular; the first scenario (dragging shapes) is one of many.
6. **Internet Access:** The system might operate offline or sync data later, so local storage and caching mechanisms should be considered.

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c. Questions to the Client

1. **Data Storage & Researcher Access**
    - Is there an existing **database infrastructure** or should we create and host a new one?
    - Will researchers **log in** to access user data, or will data be exported manually?
2. **User Identification**
    - Should each session be linked to a **researcher ID** or a **user ID**?
    - Is **authentication** (e.g., login system) required for researchers or users?
3. **Planned Scenarios**
    - What are the **future experiences** or training modules you want to include?
    - Should these scenarios focus on specific **skill sets** (e.g., hand-eye coordination, technical workflows)?
4. **Smart Assistant Scope**
    - Do you want a fully **dynamic assistant** (AI-driven) or **scripted guidance**?
    - Should the assistant adapt to user progress and mistakes?
5. **Progress Tracking**
    - What **metrics** are important to you? (e.g., time to complete task, accuracy, number of mistakes?)
    - Should we include **heatmaps**, **gaze tracking**, or **interaction logs**?
6. **User Roles**
    - Will there be **multiple user roles** (e.g., trainee, researcher, admin)?
    - Should different roles have different access levels or UIs?

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d. Proposed MVP Feature Scope

The MVP (Minimum Viable Product) will include:

1. **Virtual Lab Environment**
    - Basic virtual lab room with interactable objects.
2. **User Navigation & Interaction**
    - Move/rotate using joysticks.
    - Drag and place objects with intuitive snapping and feedback.
3. **Smart Assistant (Onboarding)**
    - Voice-guided instructions.
    - Contextual hints and tips for VR interactions.
4. **Training Scenario #1: Shape Matching**
    - 5 unique 3D shapes must be dragged into matching slots (with material variance).
5. **UI Feedback**

- On-screen UI panel indicating joystick controls and action hints.

6. **Session Tracking**
   - Save/load user progress using a researcher or user ID.
   - Store actions, time taken, and results.

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e. Architecture Overview & System Diagram

System Architecture Overview

Front-End (Unity VR App)

- XR Interaction Toolkit for interaction
- Unity UI System for instructions and menus
- Audio Manager for smart assistant guidance
- Scene Manager for training modules
- Data Manager for session save/load

Back-End (Optional Cloud Integration)

- REST API (FastAPI or Node.js) to store/retrieve session data
- Database (PostgreSQL / Firebase / DynamoDB)
- Authentication layer (if needed for researchers)

Data Format

- Each session stored with:
  - `UserID / ResearcherID`
  - `Timestamp`
  - `ScenarioName`
  - `ActionLogs[]` (time, object, action)
  - `CompletionStatus`

System Diagram

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+------------------------+
|   VR User with Headset |
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          v
+------------------------+
|   Unity VR Application |
|  - Smart Assistant     |
|  - Shape Matching Task |
|  - UI Guidance         |
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          |
          v
+------------------------+
|     Session Manager    |
| - Save/Load Progress   |
| - Attach Researcher ID |
+------------------------+
          |
          v
+------------------------+       (Optional Network Layer)
|     Local Storage      | <-----> Cloud Database / REST API
| (or Networked Storage) |
+------------------------+
```

Conclusion

The VR Lab system is being developed with scalability and user-friendliness in mind. The MVP addresses the core user interaction and training needs while allowing extensibility for advanced modules and deeper research analytics. Early involvement and feedback from the client on database integration, planned scenarios, and session tracking details are crucial to shaping a robust VR platform.